

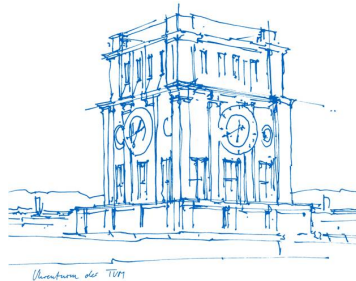
HeatPath-GNN

Learning where to place conductive particles

Thermal transport through
particle contact networks

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Motivation: conductive material is a limited resource

Heat transfer through particle beds

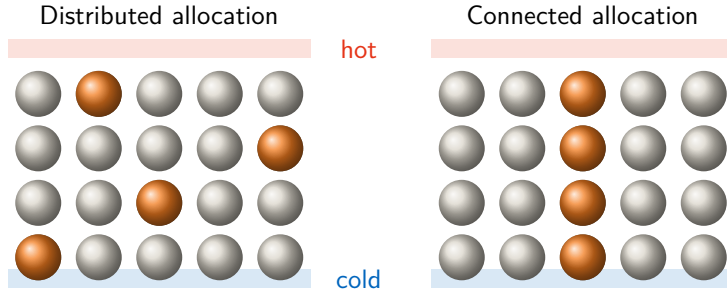
- Heating and cooling granular products
- Thermal storage and temperature control
- Transport depends on contacts between particles

A design choice

- Highly conductive material may be costly
- The material budget is fixed
- Its position within the bed can change the conducting network

Can we obtain better heat transfer from the same amount of conductive material?

The allocation problem



Orange: high conductivity. Grey: low conductivity. Schematic only.

Geometry and material amount stay fixed. Only particle labels change.

Thirty packings with the same nominal porosity

Quantity	Value
Particles	500
Diameters [mm]	1.5 / 2.0 / 2.5
Counts	100 / 300 / 100
Nominal porosity	0.38
Independent seeds	30

- Periodic boundaries in x and y
- Two fixed thermal walls in z
- No gravity during generation
- Grow small spheres, then relax
- Accept final translational $KE \leq 10^{-12}$ J

DEM: Hertz–Mindlin, $E = 50$ MPa, $\nu = 0.25$, restitution 0.30, friction 0.50.

Contact conductance links temperature difference to heat rate

Conductance: units W/K

$$Q_{ij} = G_{ij}(T_i - T_j)$$

G_{ij} : between particles i and j .

G_{iw} : between particle i and wall w .

Q_{ij} : heat rate [W].

Material conductivity k has units W/(m K).

Steady balance at particle i

$$\sum_j G_{ij}(T_i - T_j) + \sum_w G_{iw}(T_i - T_w) = 0$$

Whole-bed conductivity

$$k_{\text{eff}} = \frac{QH}{A(T_{\text{hot}} - T_{\text{cold}})}$$

Q : wall heat rate, H : height, A : wall area.

A larger conductance transfers more heat for the same temperature difference.

Thermal conductance follows contact size and material

Particle–particle contact

$$a_{ij} = \sqrt{R_{ij}^* \delta_{ij}}, \quad R_{ij}^* = \frac{R_i R_j}{R_i + R_j}$$

$$G_{ij} = \frac{4a_{ij}k_i k_j}{k_i + k_j}$$

δ_{ij} : geometric overlap.

a_{ij} : Hertz contact radius.

Ideal circular constriction model. Its small-contact assumption needs physical assessment.

Material assignment

$$k_i \in \{1, 10\} \text{ W m}^{-1} \text{ K}^{-1}$$

Isothermal wall

$$G_{iw} = 4a_{iw}k_i$$

Two spreading resistances in series give the particle-contact formula.

Every method uses the same conductive-material budget

Diameter [mm]	1.5	2.0	2.5
Total particles	100	300	100
High-conductivity particles	10	30	10

$$\max_{\mathbf{s}} k_{\text{eff}}(\mathbf{s}), \quad s_i \in \{0, 1\}, \quad \sum_{i \in \mathcal{B}_b} s_i = q_b$$

$s_i = 1$: conductive particle. $\mathcal{B}_b$: size class. q_b : class quota.

Fifty of 500 particles are conductive, with exactly 10% in each size class.

Reference methods select which 50 particles become conductive

Method	Selection rule within each size class
Random allocation	Draw 10 / 30 / 10 particles at random. Repeat 100 times and average the resulting bed conductivities.
Degree ranking	Count each particle's contacting neighbours. Select the 10 / 30 / 10 particles with the highest counts.
Heat-throughput ranking	First solve the all-low-conductivity bed. Select the 10 / 30 / 10 particles with the largest baseline heat-throughput scores.

Each allocation gives a list of high-conductivity particle IDs, then a thermal solve gives k_{eff} .

Swap search starts from the better reference allocation

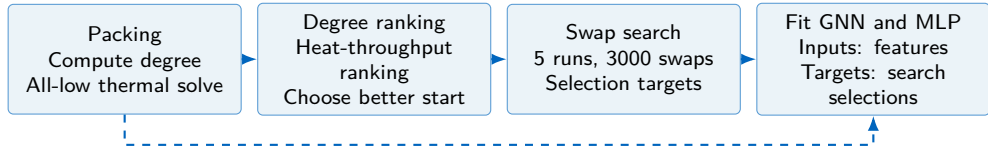
1. Evaluate the allocations from degree ranking and heat-throughput ranking. Start from whichever gives the larger k_{eff} .
2. Randomly choose a size class, then one high- k particle and one low- k particle from that class.
3. Exchange their material labels and solve the thermal network.
4. Keep the swap if k_{eff} increases. Otherwise keep the previous allocation.

Five runs use the same starting allocation and different random swap sequences.

Each run proposes 3000 swaps. Retain the best final allocation found.

Where the methods enter training, prediction and evaluation

Training: compute features and generate search targets



Particle features, including degree and heat throughput

Prediction on a new packing: features remain necessary



Study evaluation: compare GNN with random allocation, degree ranking, heat-throughput ranking, MLP and swap search on the same held-out packing.

Degree and heat throughput have three distinct roles

Role	How the quantities are used	Output
1. Initialize search	Degree ranking and heat-throughput ranking each select particles.	Better allocation starts swap search.
2. Supply features	GNN and MLP use degree and heat-throughput values during training and prediction.	Learned particle scores.
3. Compare methods	Evaluate both ranking methods on the held-out packing.	Reference k_{eff} values to compare with GNN.

Feature: a value for each particle. Ranking method: a rule for selecting particles.

All-low thermal solve before material allocation

All-low thermal solve: identical conductivity for every particle

$$k_i = k_{\text{low}} = 1 \text{ W m}^{-1} \text{ K}^{-1} \quad \text{for every particle}$$

All-low temperature $T_i^{(0)}$

The thermal solve gives each particle's temperature.

Temperatures vary between the 301 K and 300 K walls, although all particles have the same conductivity.

All-low heat-throughput score

$$s_i^{(0)} = \sum_{j \in \mathcal{N}(i)} \left| G_{ij}^{(0)} (T_i^{(0)} - T_j^{(0)}) \right|$$

Sum the absolute heat rates at particle–particle contacts [W]. A larger score indicates more contact heat traffic.

These inputs are available before we choose any high-conductivity particles.

The GNN uses geometry and the all-low thermal solution

Eight particle features

- Diameter, height and contact degree
- Hot-wall, cold-wall and anchored flags
- Normalized all-low temperature
- Normalized all-low heat-throughput score

Six contact features

- Contact radius
- Relative displacement in x, y, z
- Absolute vertical alignment
- Normalized absolute all-low contact heat rate

Three message-passing layers, width 64. MLP uses the same particle features. GNN additionally uses neighbour messages and contact features.

Training targets describe how often search selects each particle

A soft selection target

$$y_i = \frac{\text{search runs selecting particle } i}{5}$$

Example: selection in four of five runs gives $y_i = 0.8$.

The network predicts a score z_i and $p_i = \text{sigmoid}(z_i)$.

Two parts of the training loss

$$L = L_{\text{ranking}} + 0.2L_{\text{BCE}}$$

Ranking: favour particles frequently selected by search, within each size class.

Binary cross entropy (BCE): penalize disagreement between p_i and y_i .

The model learns particle priorities. Exact quotas are enforced after scoring.

Scoring assigns priorities, then ranking enforces the budget

1. Run the trained model to obtain one numerical score per particle.
A higher score means higher priority for high conductivity.
2. Convert scores to ranks within each size class, for each of three trained repeats.
3. Average those ranks and select the top 10 / 30 / 10 particles.
4. Assign $k_{\text{high}} = 10$ to those particles and solve the thermal network to verify k_{eff} .

Fifty selected particles, regardless of how many predicted probabilities exceed 0.5.

Each test packing is withheld from model fitting

28 training packings
weights and scalers

1 validation packing
early stopping

1 test packing
reported allocation

- Repeat for all 30 test packings
- Up to 400 epochs and patience 60
- Compare methods on the same held-out geometries

This tests transfer between realizations of the specified packing.

Gain and recovery answer different questions

$$\text{Gain}_m = 100 \frac{k_m - k_{\text{random}}}{k_{\text{random}}}$$

$$\text{Recovery}_m = 100 \frac{k_m - k_{\text{random}}}{k_{\text{search}} - k_{\text{random}}}$$

Gain

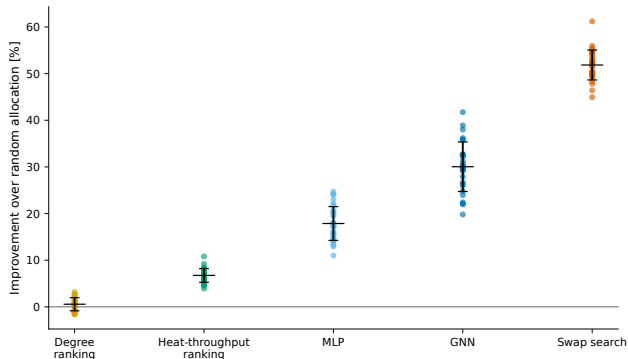
How much better is this allocation than random?

Recovery

How much of the search improvement does the model obtain?

Compute both per packing, then average. Error bars describe sample SD across packings.

GNN improves conductivity by 30% over random

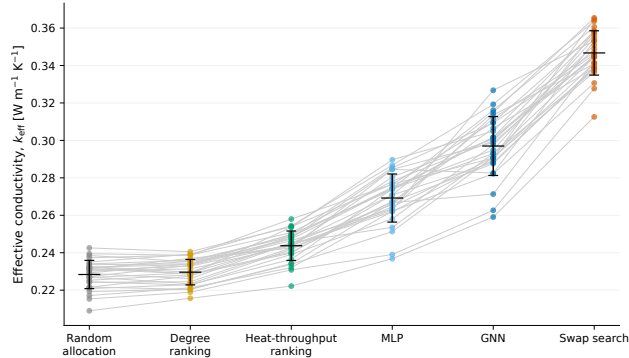


Mean improvement

Degree ranking	0.6%
Heat-throughput ranking	6.7%
MLP	17.9%
GNN	30.0%
Swap search	51.8%

Same geometry and material budget for every comparison.

Absolute conductivity remains below the search reference

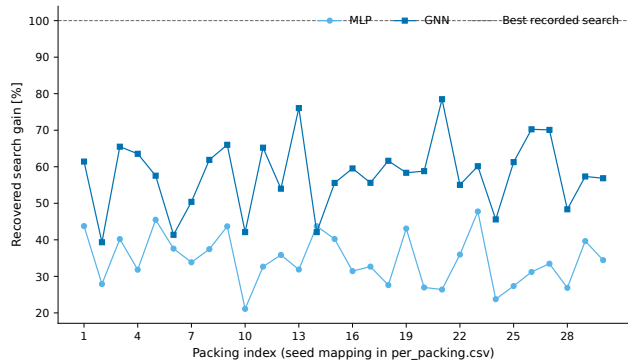


Mean k_{eff} [$\text{W}/(\text{m K})$]

Random allocation	0.2284
MLP	0.2692
GNN	0.2970
Swap search	0.3467

Grey lines connect results on the same packing.

GNN beats MLP on 29 of 30 held-out packings



Recovered search improvement

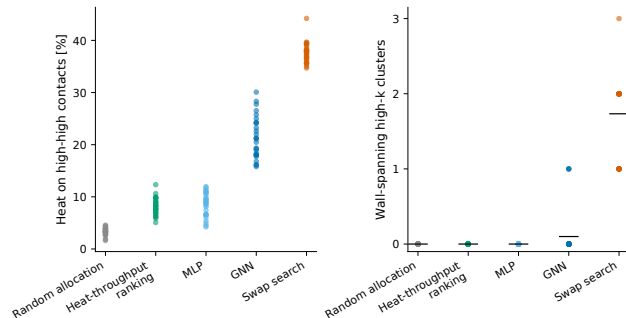
GNN: **58.0%**

SD: 9.8 percentage points

MLP: 34.5% mean

Improvement is consistent, but performance varies between packings.

Contact-network diagnostics help interpret allocation

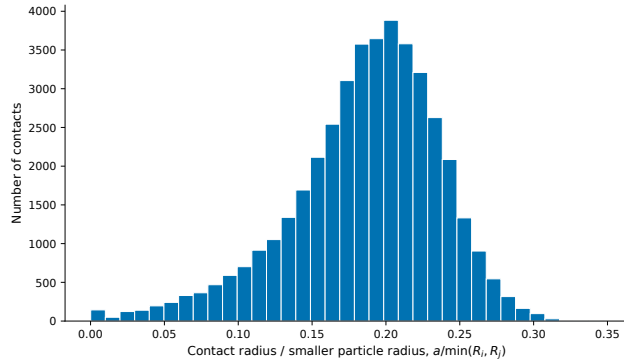


Two diagnostics

- Fraction of absolute contact heat carried by high-high contacts
- Number of high-conductivity clusters spanning both walls

These describe the selected network. They do not prove causality.

Numerical checks pass, but contact sizes need scrutiny



Recorded checks

- Maximum final KE:
 $8.4 \times 10^{-13} \text{ J}$
- Maximum energy imbalance:
 9.01×10^{-12}

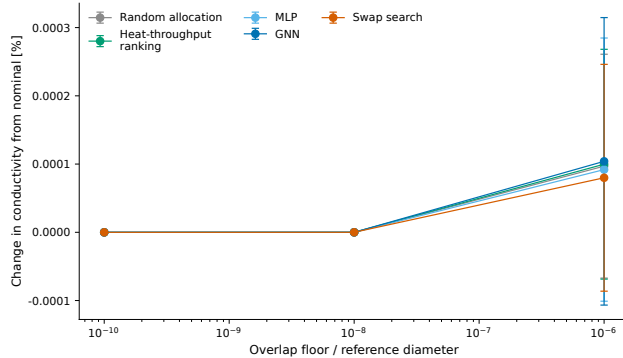
Contact radius / smaller radius

Median: 0.192

95th percentile: 0.258

Maximum: 0.347

The numerical overlap floor has little influence



Fixed allocations

- Vary the numerical overlap floor
- Largest absolute conductivity change: 0.000923%
- Sensitivity is negligible over the tested range

This does not test stiffness, loading or alternative contact laws.

Fast scoring is one part of the complete prediction procedure

What the reported times cover

- Score all particles with trained models
- Mean GNN ensemble scoring: 8.63 ms
- Mean MLP ensemble scoring: 0.295 ms

Additional work for a new packing

- Solve the all-low-conductivity baseline
- Build particle and contact features
- Combine ranks and enforce quotas
- Solve the selected thermal network

Search uses about 14,964 thermal solves per packing. Training is an additional upfront cost.

Scoring time alone is insufficient to claim an end-to-end speedup.

Conclusions and next research steps

What the campaign establishes

- Material placement strongly affects modelled conductivity
- GNN gains 30% over random allocation
- Neighbour information improves on the MLP
- 58% of best-found search gain is recovered

What remains to establish

- Physical sensitivity of contact areas
- Transfer to other packing specifications
- GNN-initialized search versus random starts
- Complete runtime and manufacturing feasibility

A learned particle ranking captures useful thermal-network structure.